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COMPUTER SECURITY CONTROL AGAINST CYBER SECURITY

Cyber Security II – Case Study submitted to PSGR Krishnammal College for Women

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TABLE OF CONTENTS

S.NO	TITLE	PAGE.NO
	ABSTRACT	I
1	INTRODUCTION	1
2	BACKGROUND AND LITERATURE REVIEW	2
3	METHODOLOGY	3
4	CASE NARRATIVE	4
5	CASE STUDY	5
6	CONCLUSION AND RECOMMENDATIONS	7
7	FUTURE RECOMMENDATIONS	8
8	REFERENCES	9

ABSTRACT

Computers and information systems have become essential components of modern society, supporting communication, business operations, education, banking, healthcare, and data management. However, the growing dependence on digital technologies has also led to an increase in cyber attacks, including malware, phishing, ransomware, denial-of-service attacks, password attacks, and unauthorized access. These attacks can result in financial losses, data breaches, operational disruptions, and reputational damage to individuals and organizations.

Computer security controls play a crucial role in protecting systems, networks, applications, and sensitive information from cyber threats. This study examines the importance of implementing effective security controls, including firewalls, antivirus and anti-malware software, access control, multi-factor authentication, encryption, intrusion detection and prevention systems, regular software updates, data backup, and employee security awareness. It also examines how preventive, detective, and corrective controls work together to reduce cyber security risks.

The study further analyzes real-world cyber attack incidents to understand the consequences of inadequate security controls and the importance of continuous security monitoring. The findings emphasize that cybersecurity cannot depend on a single security mechanism. Instead, organizations require a layered security approach that combines technological controls, organizational policies, employee awareness, regular risk assessment, and incident response procedures. Effective implementation of these controls can significantly reduce the likelihood and impact of cyber attacks and help organizations maintain confidentiality, integrity, and availability of information.

1. INTRODUCTION

The rapid development of information technology has transformed the way individuals and organizations store, process, and exchange information. Computers, networks, cloud services, mobile devices, and digital applications are now an essential part of everyday activities. Organizations depend on computer systems for financial transactions, customer management, communication, business operations, and storage of critical information.

At the same time, the increasing use of technology has created opportunities for cyber criminals to exploit weaknesses in computer systems. Cyber attacks can target individuals, organizations, government institutions, educational institutions, and financial systems. Common threats include phishing, malware, ransomware, password attacks, social engineering, denial-of-service attacks, data theft, and unauthorized system access.

Computer security refers to the practices, technologies, policies, and procedures used to protect computer systems and information from unauthorized access, modification, destruction, or disruption. Security controls are an important component of computer security because they provide mechanisms for preventing attacks, detecting suspicious activities, responding to incidents, and recovering affected systems.

Security controls can be classified into different categories. Preventive controls are designed to stop security incidents before they occur. Examples include firewalls, strong passwords, access controls, encryption, and multi-factor authentication. Detective controls identify suspicious activities or security incidents, such as intrusion detection systems, security monitoring, and log analysis. Corrective controls help organizations recover after an incident through methods such as data restoration, incident response, and system recovery.

Therefore, implementing appropriate computer security controls is essential for protecting sensitive information and maintaining reliable digital operations. A strong cybersecurity strategy requires multiple layers of protection rather than relying on a single security solution.

2. BACKGROUND AND LITERATURE REVIEW

Cybersecurity has become an important area of research due to the continuous evolution and increasing complexity of cyber threats. Earlier computer security mainly focused on protecting individual systems from viruses and unauthorized access. However, organizations now face more sophisticated threats such as automated attacks, social engineering, ransomware, supply-chain attacks, and exploitation of software vulnerabilities Information security is commonly based on the CIA triad: Confidentiality, Integrity, and Availability. Confidentiality ensures that information is accessible only to authorized users, integrity protects information from unauthorized modification, and availability ensures that systems and data remain accessible when needed. These three principles form the foundation of effective information security.

Cybersecurity frameworks emphasize the use of multiple security controls to protect systems and information. Firewalls monitor and control network traffic, while antivirus and endpoint protection help detect and prevent malicious software. Access control mechanisms ensure that users can access only the resources for which they have permission.

Authentication is another important security measure. Password-based systems can be affected by stolen credentials, brute-force attacks, and credential stuffing. Multi-factor authentication (MFA) provides an additional layer of protection by requiring users to verify their identity through more than one authentication method. Encryption is widely used to protect sensitive information by converting readable data into an encoded form. It helps secure both stored information and data transmitted across networks, reducing the risk of unauthorized access.

Employee awareness is also essential because human actions can contribute to cyber incidents. Phishing emails, malicious attachments, fake websites, and social engineering techniques can trick users into revealing sensitive information or credentials. Regular cybersecurity awareness and training programs can help reduce these risks.

Modern cybersecurity practices also emphasize continuous monitoring, risk assessment, incident response, and recovery. Organizations must regularly evaluate their security controls and update them according to emerging threats. Therefore, cybersecurity is an ongoing process that requires a combination of technology, policies, employee awareness, and regular security improvements.

3. METHODOLOGY

This study adopts a descriptive and analytical approach to examine the importance of computer security controls in protecting systems against cyber attacks.

Secondary data was collected from cybersecurity research papers, academic publications, security frameworks, case studies, industry reports, and trusted online sources. The information was reviewed to identify common cyber threats, vulnerabilities, security controls, and effective protection mechanisms.

The study focuses on major security controls, including:

- Firewalls
- Antivirus and anti-malware software
- Access control and multi-factor authentication
- Encryption
- Intrusion Detection and Prevention Systems (IDS/IPS)
- Software patching and updates
- Data backup and recovery
- Security awareness and employee training
- Security monitoring and incident response

Real-world cyber attack incidents were also examined to understand how security weaknesses can be exploited and how effective controls can reduce their impact.

The findings were synthesized to explain how organizations can implement a layered security strategy by combining technical controls, employee awareness, organizational policies, and continuous monitoring.

4. CASE NARRATIVE

A significant example demonstrating the importance of computer security controls is the WannaCry ransomware attack of 2017.

WannaCry was a ransomware attack that affected computer systems across numerous countries and organizations. The malware encrypted files on infected systems and demanded payment from victims in exchange for restoring access to their data. One of the major factors associated with the spread of the attack was the exploitation of a vulnerability in the Windows operating system. Systems that had not applied the relevant security updates were particularly exposed.

The incident demonstrated the importance of regular software patching and vulnerability management. Organizations that failed to update vulnerable systems faced a greater risk of infection and operational disruption.

The attack also demonstrated the importance of network segmentation, endpoint protection, data backups, access controls, and incident response procedures. If one system becomes infected, properly segmented networks can limit the ability of malware to spread throughout an organization.

Organizations can reduce the impact of ransomware by maintaining regular offline or protected backups. In the event of an attack, reliable backups can allow affected systems and data to be restored without depending entirely on attackers.

The WannaCry incident therefore highlights an important principle of cybersecurity: organizations must continuously identify vulnerabilities and apply appropriate security controls before attackers can exploit them.

5. CASE STUDY

The analysis of computer security practices shows that cyber attacks can occur when organizations fail to implement appropriate preventive, detective, and corrective controls. One major weakness is the failure to update operating systems and applications regularly. Software vulnerabilities can provide attackers with opportunities to gain unauthorized access. Regular patch management can significantly reduce exposure to known vulnerabilities.

Another major risk is weak authentication. Password reuse, simple passwords, and stolen credentials can allow attackers to gain access to systems. Multi-factor authentication provides an additional layer of protection by requiring users to verify their identity through multiple factors. Network security controls are also important.

Firewalls can monitor incoming and outgoing network traffic and prevent unauthorized connections. Intrusion Detection and Prevention Systems can identify suspicious network activities and, in some cases, automatically block malicious traffic.

Endpoint security provides another layer of protection. Antivirus and anti-malware solutions can detect malicious files and suspicious behavior on computers and other devices.

Data encryption is particularly important for protecting sensitive information. Even if attackers gain access to encrypted data, the information may remain difficult to use without the required decryption key.

Regular backups are essential for recovery from ransomware, accidental deletion, hardware failure, or other destructive incidents. However, backups should be protected from unauthorized access and ransomware so that attackers cannot easily encrypt or delete them.

Human awareness is another critical component. Employees should be trained to recognize phishing emails, suspicious links, malicious attachments, fake login pages, and social engineering attempts.

The following security controls can therefore be considered as part of a layered cybersecurity strategy:

Table 5.1. Major Computer Security Controls and Their Purposes

Security Control	Purpose
Firewall	Controls and filters network traffic
Antivirus/Anti-malware	Detects and removes malicious software
Access Control	Restricts unauthorized access
Multi-Factor Authentication	Adds additional identity verification
Encryption	Protects sensitive information
IDS/IPS	Detects and prevents suspicious network activity
Software Updates	Removes known vulnerabilities
Backup	Enables recovery after data loss or ransomware
Security Awareness	Reduces human-related security risks
Incident Response	Helps organizations respond to attacks
Security Monitoring	Identifies suspicious activity continuously

Table 5.1 Illustrates the major security controls commonly used to protect computer systems and networks from cyber threats. It highlights the primary purpose of each control and shows how different measures contribute to preventing, detecting, and responding to cyber attacks.

Effective cybersecurity requires these controls to work together. A firewall alone cannot protect an organization from phishing, stolen credentials, or malicious insiders. Similarly, antivirus software cannot prevent every type of network attack. A layered approach therefore provides stronger protection.

Organizations should also regularly conduct risk assessments to identify vulnerabilities and determine which security controls are most appropriate. Security policies should be reviewed periodically because cyber threats continuously evolve.

6. CONCLUSION AND RECOMMENDATIONS

In the modern digital environment, computer systems face threats such as malware, phishing, ransomware, unauthorized access, and network attacks. These threats can cause financial, operational, and reputational damage.

The study shows that security controls such as firewalls, encryption, access control, multi-factor authentication, IDS/IPS, backups, and software updates are essential for reducing cybersecurity risks. Preventive controls help stop attacks, detective controls identify suspicious activities, and corrective controls support recovery.

Organizations should follow a defense-in-depth approach by combining technology, security policies, employee awareness, and continuous monitoring.

Recommendations

1. Keep software and systems updated.
2. Use strong passwords and MFA.
3. Configure and monitor firewalls.
4. Maintain antivirus and endpoint protection.
5. Encrypt sensitive data.
6. Maintain secure backups.
7. Provide employee security training.
8. Conduct regular security audits.
9. Monitor networks continuously.
10. Maintain an incident response plan.

These measures can improve an organization's ability to prevent, detect, respond to, and recover from cyber attacks.

7. FUTURE RECOMMENDATIONS

In the future, organizations should move from traditional security approaches toward more proactive and intelligent cybersecurity strategies. Artificial intelligence and machine learning can be used to identify unusual behavior, detect potential threats, and support faster security responses.

Organizations should also adopt continuous security monitoring rather than relying only on periodic security assessments. Automated monitoring systems can analyze network traffic, user behavior, and system logs to identify suspicious activities.

The use of Zero Trust security principles can further improve protection by requiring continuous verification of users, devices, and access requests rather than automatically trusting users inside an organization's network.

Cloud computing and remote work also require organizations to strengthen endpoint security, identity management, and secure access controls. As more business operations move to cloud environments, organizations should regularly assess cloud configurations and permissions.

Future cybersecurity programs should also focus strongly on employee education. Regular phishing simulations, security awareness programs, and practical training can help employees recognize emerging threats.

Governments and organizations should continue developing cybersecurity policies, standards, and regulations to improve data protection and incident reporting. Collaboration between organizations, cybersecurity professionals, governments, and technology providers can help identify emerging threats and develop effective countermeasures.

By combining artificial intelligence, continuous monitoring, Zero Trust principles, strong security controls, employee awareness, and effective cybersecurity policies, organizations can build a more resilient digital environment capable of responding to increasingly sophisticated cyber attacks.

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